MUHAMMAD HISYAM

SOFTWARE ENGINEERING PROJECT SHOWCASE

Full Stack, Desktop Cross-Platform & AI Deep Learning Applications

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CORE TECHNICAL CAPABILITIES

Languages	C#, TypeScript, JavaScript, Rust, Python, Golang, Java, PHP, PL/pgSQL, HTML5, CSS3
Frontend & Desktop	Tauri (Rust Runtime), Astro 5, React.js, SolidJS, SvelteKit, Next.js, Vue.js, Tailwind CSS 4
Backend & Cloud	.NET Core, Supabase, Node.js, Express.js, Go (Gin/Fiber), Java Spring Boot, Laravel, WebSockets
AI & Deep Learning	PyTorch, U-Net Architecture, OpenCV, Model Context Protocol (MCP), Google Gemini API
Database & Infrastructure	PostgreSQL, Supabase RLS, MongoDB, Redis, Docker, Git, Vite, Bun, Biome, Linux Shell

FEATURED SYSTEM SHOWCASE & CASE STUDIES

1. RetailHub – Point of Sale & Inventory Management Platform [GitHub Repo]

Technologies: Tauri · Rust · TypeScript · TanStack Router · Supabase · PostgreSQL · Tailwind CSS

Architecture & Key Features:

- High-performance desktop and cross-platform cashier POS application engineered with Tauri (Rust native backend runtime) and TypeScript frontend.
- Multi-tenant database design using Supabase PostgreSQL, incorporating Row Level Security (RLS) policies and PL/pgSQL stored routines.
- Real-time inventory sync across multiple checkout terminals using Supabase WebSockets channels and instant offline-first local caching mechanisms.

2. Interactive Developer Portfolio & Dev Suite Arcade [GitHub Repo] [Live Site]

Technologies: Astro 5 · SolidJS · Tailwind CSS 4 · TypeScript · HTML5 Canvas

Architecture & Key Features:

- Built using Astro 5 zero-JS island architecture combined with SolidJS fine-grained reactive primitives.
- Integrated an embedded Unix terminal shell (`hisyam-cli`), REST API request inspector demo, background audio player, and **8 playable mini-games** (2048, Whack-a-Bug, Sudoku, Pong, Minesweeper, Snake, Memory Match, Typing Speed).
- Automated telemetry pipeline fetching real-time GitHub repository data and build diagnostics.

3. Pneumonia U-Net Medical Image Segmentation [GitHub Repo]

Technologies: Python PyTorch OpenCV Torchvision Matplotlib

Architecture & Key Features:

- Deep learning medical computer vision pipeline utilizing PyTorch and U-Net encoder-decoder network architecture.
- Automated mask generation for chest X-ray scans to isolate and segment pulmonary infection regions accurately.
- Evaluated against medical benchmark metrics achieving high Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) scores.

4. Gemini Model Context Protocol (MCP) Client [GitHub Repo]

Technologies: TypeScript Node.js Google Gemini API MCP Protocol

Architecture & Key Features:

- Agentic AI integration client implementing Anthropic's open Model Context Protocol (MCP) specification for Google Gemini models.
- Enables local system tool discovery, structured JSON function calling, and automated agent workflow execution.

5. Real-Time Energy Telemetry & Sensor Analytics [GitHub Repo]

Technologies: TypeScript React.js Node.js WebSockets Chart.js

Architecture & Key Features:

- IoT energy consumption analytics dashboard processing high-frequency live sensor streams over WebSockets.
- Features custom time-series data visualization, threshold alert notifications, and automated reconnection fallback mechanisms.

6 & 7. Microservices & Distributed Concurrent Backend Systems

- **Golang & SvelteKit Real-Time Chat App** [GitHub Repo]
  Go Goroutines WebSockets SvelteKit
  Concurrent multi-room chat server powered by Golang channels paired with a reactive SvelteKit frontend.
- **Secure Java JWT Authentication Microservice** [GitHub Repo]
  Java Spring Boot JWT PostgreSQL
  Enterprise authentication service featuring sliding refresh tokens and version-based session invalidation.